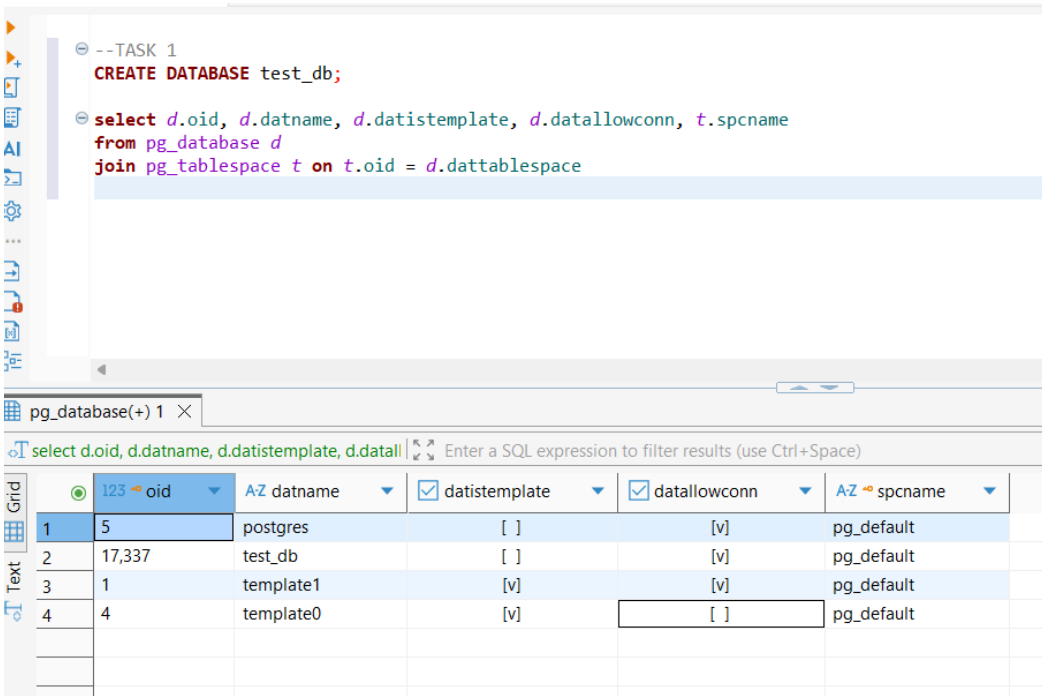
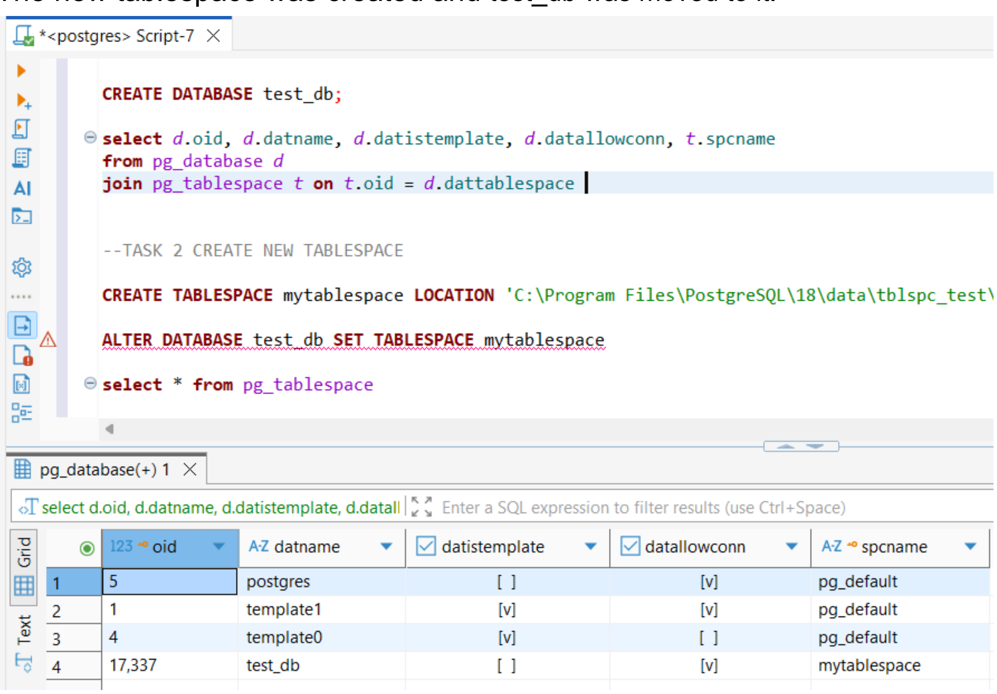
--TASK 1 CREATE NEW DATABASE

The test_db was created to the default tablespace pg_default.



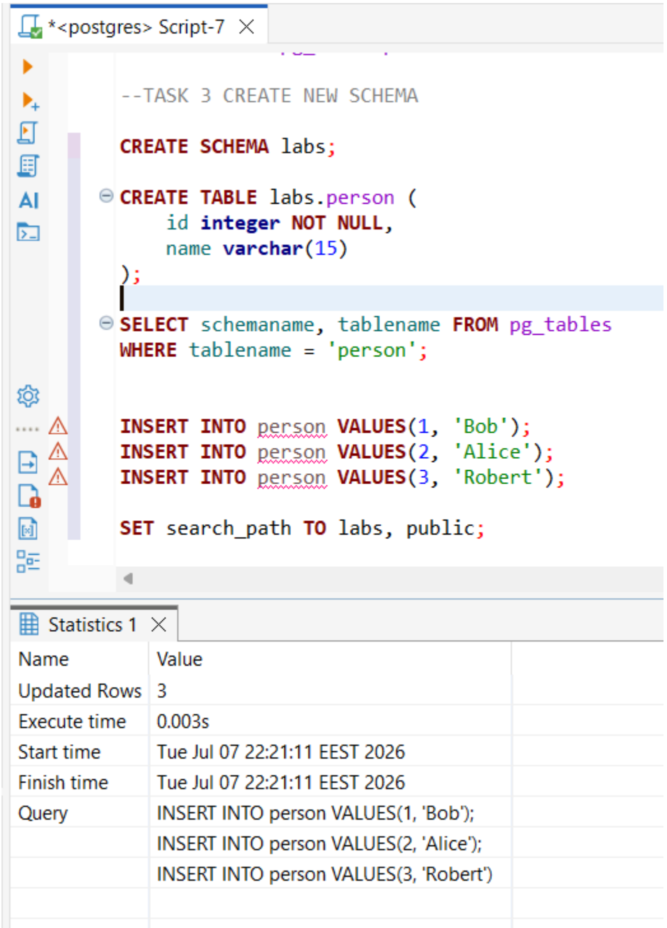
--TASK 2 CREATE NEW TABLESPACE

The new tablespace was created and test_db was moved to it.



--TASK 3 CREATE NEW SCHEMA

The schema ‘lab’ was created and the search route was directed to it.
Initial data inserted.



-- TASK 4 INVESTIGATE MVCC

Separate transaction blocks to insert, update, and delete rows were performed
The transaction flow and records positions were reflected on t_xmin and t_xmax values.

person 1 | person 2 | Results 3 | Results 5 | person 5 | Statistics 5 | Results 6 | Results 8

SELECT t_xmin, t_xmax, t_ctid, tuple_data_split('la | Enter a SQL expression to filter results (use Ctrl+Space)

	t_xmin	t_xmax	t_ctid	tuple_data_split
1	954	0	(0,1)	\x01000000 [+1]
2	954	956	(0,5)	\x02000000 [+1]
3	954	957	(0,3)	\x03000000 [+1]
4	955	0	(0,4)	\x04000000 [+1]
5	956	0	(0,5)	\x02000000 [+1]
6	958	959	(0,6)	\xe7030000 [+1]

--TASK 5 INVESTIGATE VACUUM

VACUUM labs.person command identified the 3 dead row versions and marked them reusable. The 'Sarah' record landed on the 'new' row.
VACUUM FULL labs.person command built a completely empty data file and copied *only* the live rows ("Bob", "John", updated "Alex" and "Sarah") over to it. The old file was deleted.

erson 1 | person 2 | Results 3 | Results 5 | person 5 | Results 6

ELECT t_xmin, t_xmax, t_ctid, tuple_data_split('la | Enter a SQL expression to filter results (use Ctrl

	t_xmin	t_xmax	t_ctid	tuple_data_split
1	954	0	(0,1)	\x01000000 [+1]
2	960	0	(0,2)	\x05000000 [+1]
3	955	0	(0,3)	\x04000000 [+1]
4	956	0	(0,4)	\x02000000 [+1]